

AI-BASED SOCIAL ENGINEERING DETECTOR

Idea Evolution After Initial Research

1. ORIGINAL IDEA

AI-Based Social Engineering Detector

Concept:

An AI system that analyzes emails/messages and detects whether they are examples of social engineering or scams.

Basic flow:

Message → AI/ML Model → Scam / Safe

Possible technologies:

- NLP
- BERT / Transformers
- Machine Learning
- Intent Detection
- Text Classification

Problem:

Basic AI-based phishing/scam detection already exists, so this version would have **limited novelty**.

2. WHAT OUR RESEARCH FOUND

Recent research shows that social-engineering detection is moving beyond simple message classification.

Important finding:

A 2026 study introduced **COVA**, a dataset of 3,201 multi-turn scam conversations, specifically to study conversational smishing rather than individual messages.

A newer version, **COVA-X**, expanded this to **10,985 multi-turn conversations** and showed that Longformer could outperform traditional XGBoost when enough conversational data was available.

This confirms that **multi-turn conversational social-engineering detection is an active and emerging research area**.

[Research Paper — COVA-X: An Expanded Synthetic Conversation Dataset for Multi-Turn Smishing Detection](#)

3. CHANGE IN OUR IDEA

Instead of:

"Detect whether a message is a scam."

We want to investigate:

"Understand how a social-engineering attack develops throughout a conversation."

New Concept:

AI-Based Multi-Stage Conversational Social Engineering Detector

4. WHAT WILL BE DIFFERENT?

A. Conversation-Level Detection

Instead of analyzing one message:

Message → Scam / Safe

Analyze the entire conversation:

Message 1

↓

Message 2

↓

Message 3

↓

Message 4

↓

Conversation-Level Analysis

B. Attack Progression Detection

The system could identify the stages of an attack:

Trust Building

↓

Impersonation

↓

Fear / Urgency



Information Request



Credential / OTP / Payment Request

C. Manipulation Tactic Detection

Instead of only saying "Scam", the system explains **how the attacker is manipulating the victim**.

Possible tactics:

- Authority impersonation
 - Fear
 - Urgency
 - Trust manipulation
 - Threats
 - Emotional pressure
 - Reward/scarcity
 - Credential harvesting
-

D. Explainable Risk Score

Instead of:

Scam: 93%

The system could provide:

Risk: 93/100

Reasons:

- ✓ Authority impersonation
- ✓ Urgency
- ✓ Fear
- ✓ Suspicious link
- ✓ OTP request

Current Stage:

Credential Harvesting

E. Risk Progression

The system could track how the risk changes during the conversation:

Message 1 → 12%

Message 2 → 27%

Message 3 → 51%

Message 4 → 78%

Message 5 → 94%

This allows the system to identify when a conversation starts becoming dangerous.

5. POSSIBLE RESEARCH QUESTION

Can conversation-level and context-aware AI models detect and explain the progression of multi-stage social-engineering attacks more effectively than traditional single-message classification?

Possible experiments:

Traditional ML

vs

BERT / Transformers

vs

Long-context Transformer

vs

LLM-based approach

vs

Hybrid approach

Metrics:

- Accuracy
- Precision
- Recall

- F1-score
 - False-positive rate
 - Processing time
-

6. CURRENT NOVELTY POSITION

We should NOT claim:

"Nobody has built an AI social-engineering detector."

That would be incorrect.

Existing research already covers:

- Phishing detection
- Social-engineering detection
- Multi-turn smishing detection
- NLP-based detection

For example, a 2026 systematic review examines ML techniques for detecting social-engineering attacks in text-based communication and identifies trends, limitations and research gaps.

[Systematic Review — ML Techniques for Social Engineering Detection](#)

Our potential contribution

The proposed focus is:

Detection + understanding of attack progression + manipulation tactics + explainable risk assessment

rather than simply:

Scam / Not Scam

7. OTHER RELEVANT RESEARCH

LLMs + Phishing Detection

A 2026 systematic literature review examined **36 studies** on the use of LLMs for phishing generation and detection.

It found that LLMs are increasingly being used for phishing detection, but also make phishing attacks more convincing and scalable.

[Systematic Review — LLMs in Phishing Attack Generation & Detection](#)

COVA-X — Multi-Turn Smishing

10,985 conversational examples were used to evaluate multi-turn smishing detection.

Longformer achieved approximately **79.7% accuracy and 0.779 macro-F1**, demonstrating the importance of models capable of handling longer conversational context.

[Research Paper — COVA-X](#)

Broader Phishing Research Review

A 2026 review discusses modern phishing detection approaches and how AI is changing the phishing landscape.

[Review — Phishing in the Age of Distributed Intelligence](#)

8. FINAL EVOLUTION

BEFORE

AI Social Engineering Detector

Message



ML/NLP



Scam / Safe

AFTER RESEARCH

AI-Based Multi-Stage Conversational Social Engineering Detector

Complete Conversation



Context Analysis



Attack Type



Manipulation Tactics



Attack Progression



Risk Score



Explainable Warning



Recommended Action

9. CURRENT STATUS

Idea: Strong candidate

Research area: Active / emerging

Novelty: Needs to be validated further

Main novelty direction:

Conversation + attack progression + manipulation tactics + explainability

Next step:

Before finalizing the project, investigate:

1. Existing systems
2. Existing research papers
3. Available datasets
4. Existing GitHub implementations
5. Exact research gap
6. Feasibility
7. Possible experimental contribution

Only after this should the final project scope be selected.